

LESSON 4.10

USING CIRCUMFERENCE AND AREA TO SOLVE PROBLEMS



LESSON INTRODUCTION

MA.7.GR.1.3 Explore the proportional relationship between circumferences and diameters of circles. Apply a formula for the circumference of a circle to solve mathematical and real-world problems.

MA.7.GR.1.4 Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems.

LEARNING TARGETS

- I can use the circumference and area of a circle to solve problems.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.

ESSENTIAL QUESTIONS

- How do you find the area of a circle in a real-world problem?
- How do you find the circumference of a circle in a real-world problem?

LESSON NARRATIVE

Students have previously derived the formulas for determining the circumference and the area of a circle. This lesson develops procedural fluency with the formulas using real-world problems.

COMPONENT SUMMARY

4.10.1 WARM-UP (~5 MIN.)

Compare the circumferences of circles to the perimeters of rectangles

4.10.3 YOUR TURN! (15 MIN.)

Individually practice solving real-world area and circumference problems

4.10.2 ACTIVITY (20 MIN.)

Collaboratively practice solving real-world area and circumference problems

4.10.4 WRAP-UP (~5 MIN.)

Reflection on mastery of lesson learning targets

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Chart paper & markers
- Calculator

ADDITIONAL PREPARATION

- Consider modifying the people, places, and things in the questions to connect to students' interests.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle with entering the values into the calculator.
- Students may forget to determine the radius when given the diameter.

THINGS TO CONSIDER

- Consider having students create an anchor chart or Frayer Model with the formulas for area and circumference. Include examples and visuals.
- In problems that require several steps, students should not round until the final step. Rounding mid-step could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common careless errors that many students make.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

In 4.10.2 Activity, prompt students to reflect on and discuss the relationships between the radius, diameter, area, and circumference of circles and how these are applied in the solving of real-world problems.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

The 4.10.2 Activity provides an opportunity for students to reflect on and discuss how to find the area and circumference of a circle in real-world problems. Prompt students to analyze the mathematical thinking of their peers with questions like “Do you agree?” “Will this method always work?” Encourage students to explain how they obtained their answers.

DIFFERENTIATE INSTRUCTION

In 4.10.3: Your Turn! consider providing an auditory entry point by reading the problems aloud or having students use the text reader in the online version of the lesson. Encourage students to underline key information they will need to solve the problems. Then, prompt students to label the diagrams with the dimensions they know and variables for the unknown quantities. Prompt students to reference their anchor charts or Frayer Models with formulas, visuals, and examples.

4.10.1 WARM-UP: WOULD YOU RATHER?

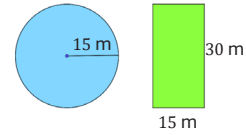
TEACHER MOVES

Ask students if they recall the formulas $C=2\pi r$ and $P=2l+2w$. Display the formulas for students to reference.



4.10.1 Warm-Up: Would You Rather?

A circle with a radius of 15 meters (m) and a rectangle with a length of 30 m and width of 15 m are shown.



1. Explain if you would rather run the circumference of the circle or the perimeter of the rectangle.

Answers will vary. The circumference of the circle is 94.2 m. The perimeter of the rectangle is 90 m. If a student wants to run the shorter distance, they will select the circle. If a student wants to run the longer distance, they will select the rectangle.



4.10.2 Activity: Sage and Scribe

For the first problem, decide who will be the Sage and who will be the Scribe. While working a problem, the Sage is the only one who can talk. The Scribe must write down only what the Sage says to solve the problem. For each problem, swap who is the Sage and who is the Scribe.

Problem A	
Name of Sage:	A lawn sprinkler sprays water in a circle with a radius of 6.27 feet (ft).
	
Name of Scribe:	How much of the lawn will be watered from the sprinkler? Use $\frac{22}{7}$ for π . 123.555 ft²
Problem B	
Name of Sage:	Felicia has a circular rug that measures $5\frac{7}{12}$ feet (ft) from the center to the edge of the rug.
	
Name of Scribe:	Felicia needs to replace the fringe on the edge of the rug after her dog chewed it. How much fringe does Felicia need to buy? Use 3.14 for π . 35.063 ft

4.10.2 ACTIVITY: SAGE AND SCRIBE

TEACHER MOVES

Students each taking on a role within the activity facilitates an equitable learning experience for both students.

- Sage: can only tell the Scribe what to write
- Scribe: can only write what is said by the Sage

DIFFERENTIATE INSTRUCTION

Make space for student agency, allowing students to do their own research on local examples in their personal lived worlds to customize their learning experiences.

Make this activity into an extension project in which students must find an example in their world and create their own problem to trade with a partner.

Provide a computer, phone, or tablet for students to research local examples and substitute for any of the examples provided.

Prompt students to bring in objects from around the house to trade with a partner and find area and circumference.

ENRICHMENT

Did your partner solve the problems in the same way you would have? Which strategy or method is the most efficient? Do the strategies or methods result in the same answer? How did you determine whether to calculate the area or circumference?

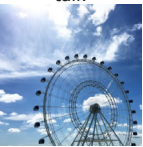
4.10.2 ACTIVITY: SAGE AND SCRIBE (CONTINUED)

TEACHER MOVES

Compare and Connect

Prompt students to reflect on and discuss the relationships between the radius, diameter, area, and circumference of circles and how these are applied in the solving of real-world problems.

How do we know that 400 feet is the diameter? What does this have to do with finding how far someone traveled around?

Problem C Name of Sage: Name of Scribe:	The Wheel at ICON Park is approximately 400 feet (ft) tall.  If a person takes one trip around the Ferris wheel, about how far will the person have traveled? Use $\frac{22}{7}$ for π. 1257.143 ft
Problem D Name of Sage: Name of Scribe:	William is painting a large image of the full moon in the image below on his wall. He wants the painting of the full moon to cover his wall from the floor to the ceiling.  If the wall is 3.048 meters (m) tall, how much of the wall will be covered in paint? Use 3.14 for π. 7.293 m²



4.10.3 Your Turn!

1. Yvonne’s company wants to put their logo on frisbees to give away at a convention.



If each frisbee has a diameter of $12\frac{1}{2}$ inches (in.), and the logo can cover the entire top of a frisbee, how much space is available for the logo on each frisbee? Use $\frac{22}{7}$ for π .
122.768 in²

2. A jeweler is asked to wrap a rare coin with gold braid to make a pendant.



If the coin has a diameter of 0.75 inches (in.), what is the length of the gold braid the jeweler will need? Use $\frac{22}{7}$ for π .
2.357 in.

4.10.3 YOUR TURN!

TEACHER MOVES

Three Reads

Consider incorporating structure with the Three Reads routine, providing scaffolding for contextualizing and decoding word problems.

- First Read: What is the question asking?
- Second Read: What important information might contribute to answering the question?
- Third Read: What will my approach be to solving? OR: What will my answer look like if it is reasonable?



4.10.4 Wrap-Up: Reflection

Today I deserve 	<i>Answers will vary.</i>
for ...	
Today I 	<i>Answers will vary.</i>
by doing ...	
Today I was 	<i>Answers will vary.</i>
by ...	

4.10.4 WRAP-UP: REFLECTION

ENRICHMENT

Consider providing question stems or sentence frames for students to enhance the details of their responses.

- How confident do you feel with the following?
- Determining whether the problem involves finding the area or circumference
- Calculating the areas of circles
- Calculating the circumferences of circles
- Rounding calculations to the nearest thousandth
- Explaining the relationship between the radius, diameter, circumference, and area of a circle